The Radar Museum of Leonardo

A unique corporate museum, a living project from the industrial heritage to the current technological innovation

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***Abstract*— Dedicated to Carlo Calosi, a pioneer of the Italian radar technology, the Leonardo Radar Museum opened on the 2nd of April 2009 in the company's site of Fusaro near Naples, for representing development in radar technology and related industrial products during the past century. The first Corporate Museum in Europe, the Radar Museum has the major aim of collecting, reorganizing and keeping alive history of radar manufacturing, for these reasons this exhibition has been located at Fusaro, one of the company plants which played an important role during this long path. The historic timeline is based on plenty of images, writings, documents, and products showing all the main steps for the radar's design and production process. This huge collection allowed to set up a centre of reference for radar manufacturers and experts, witnessing the level of excellence achieved by the Italian industry in the fields of Air Traffic Control, land and naval radars. The Radar Museum covers an area of 750 square meters, divided into four zones that include an exhibition area, a library, a simulation area and a conference room. Some of the most significant antennas realized by the Company in the past are also displayed outside the main building. The Museum currently represents lively and dynamic project.**

Keywords*—*Radar, Museum, radar technologies, Leonardo Company

I. THE RADAR HISTORY IN ITALY

The history of the Italian radar began with a famous memorandum presented by Marconi to the "Institute of Radio Engineers" on the 20 June 1922. [1]. Marconi formulated a concrete analysis about the possibility of utilizing radio waves to build an auxiliary navigation system, based on the ability of electrical waves to be reflected by conducting bodies. From this time on, the first experiments began in Britain and the United States and, even if based on different procedures, they represent the starting points that brought to the construction of equipment for radio detection, the real ancestor of the radar. The Second World War gave a crucial impulse to the development of radar that, in only five years, assumed the aspect and operational functions that remained basically unchanged till today. In 1950, the first Italian radar company was established in Baia near Naples, where the Royal Italian Torpedo Factory had been since 1921. In the same place, Microlambda was established, manufacturing radar systems under licence of the US company Raytheon. In 1960, thanks to the agreement between Microlambda and Sindel, the company Selenia was funded, and quickly established its reputation for the technical excellence of its own radar systems, which were produced and sold globally [2]. Currently, Italian industries cover almost all the radar product families and all the market needs. In industries can achieve experience from operating their systems in real environments. A coordination among Institutions, Universities and Industries is also necessary to address new Research and Development activities in innovative and challenging directions, thus maintaining the edge of the radar system market.

II. WHY A RADAR MUSEUM

According to the ICOM (International Council of Museums): "A museum is a not-for-profit, permanent institution in the service of society that researches, collects, conserves, interprets and exhibits tangible and intangible heritage. Open to the public, accessible and inclusive, museums foster diversity and sustainability. They operate and communicate ethically, professionally and with the participation of communities, offering varied experiences for education, enjoyment, reflection and knowledge sharing". This definition is now a reference in the international community. RADAR (RAdio Detection and Ranging) is a typical example of technology that developed later than its original physics. To give an idea, Maxwell described the electromagnetic spectrum in 1873, while in 1887 the German physicist Heinrich Hertz generated and detected radio waves in laboratory and at the end of the century Guglielmo Marconi carried out first transmission tests over long distances. In the same years, the German inventor Christian Hülsmeyer was the first to use radio waves to detect "the presence of distant metallic objects". In 1904 he demonstrated the feasibility of detecting a ship in dense fog, but not its distance from the transmitter, obtaining a patent for his detection device [3]. Anyway, radar became operational many years later, during the Second World War and from then on it is continuously developing, with new technologies and applications [4]. Defence, Meteorology, Transport, Astronomy, Geophysics, Oceanography are just some of the application areas showing how radars [5] are now integral part of them, also enlarging to new sectors like security for high visibility events like in the last Olympic games. In fact, during the 2012 London Olympic security assets also included surveillance and tracking radars for air defence, as well as weather radars for nowcasting. Then a

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radar museum can represent both a repository and an engine at the same time. Currently, many radar museums are part of war museums, in order to demonstrate the essential role played by radar during the Second World War and in subsequent conflicts. Therefore, they are based in sites of historical interest and in some cases under the Armed Forces management [6] [7].

III. The Radar Museum of Leonardo

The Leonardo Radar Museum is a corporate museum established in Italy in 2009 to represent how technology development, company business development and national industrial policies have walked together for over 50 years. Dedicated to Carlo Calosi [8], a pioneer of the Italian radar technology, the Radar Museum opened on the 2nd of April 2009. The company decision to create a Radar Museum derived from the awareness that most part of the industrial culture of our age lies within enterprises. The first Corporate Museum in Europe, the radar museum is based on the principles of collecting, reorganizing, and keeping alive the history of radar manufacturing, which began about 60 years ago in the Fusaro site (Naples) in Italy. The historic timeline using images, writings, documents, and products showing all the main steps for the radar's design and production process, has allowed the creation of a centre of reference on a national scale to provide imagery from radars and the manufacturers themselves, beyond technical and technological development levels achieved by Italian industry, and, by Campania industry (Figure 1). In addition to presenting all technological and innovative steps in this sector, the Radar Museum's aim is to broaden awareness about the scientific and industrial research fields in electronics and new technologies [9]. Even if born as a private initiative the museum can be considered as a reference and a meeting point for specific sector events. In fact, it can host didactic and scientific activities, lessons and conferences, providing support to experts looking for useful information for research and studies, as well as to students for completing their training. The Museum is now a lively and dynamic project that reflects the state-of-the-art of the Italian radar industry.

Fig 1. - A wide angle view of the Radar Museum © by Leonardo.

IV. The Museum

The Radar Museum covers an area of 750 sqm divided into four zones with an exhibition area, a library, a simulation area and a conference room. Some of the most significant antennas realized by the Company in the past are also showcased outside. Entering the museum, visitors can proceed along paths where a series of panels and photographs illustrates the history of radar and of the company activities in radar design and production. The simulation room is equipped with three different consoles, for ground surveillance, naval defence and air traffic control respectively, each one configured to display a different working scenario. These three sectors reflect the main business areas, where the company has gained an international leadership with more than 150 customers in the world. The museum also includes a multimedia consultation area and a library, containing historical volumes about radars and their technologies, as well as documents covering the history of the company.

Finally, the conference room can accommodate up to 50 people. Inside the Museum (Figure 2), visitors can follow a journey through a series of panels that show documents and personalities at the heart of the radar's history. In addition, the area also contains the main instruments, and the production parts developed by the Company, designed and realized by the in-house foundry.

Fig 2. - The Radar Museum 3D layout © by Leonardo

The exhibition area contains both models of past and current radars and display cases showcasing technologies and component parts (Figure 3). These cases are organized according to a series of technological paths addressing:

- Radio frequency power devices
- High Frequency devices for generation and processing of radar signals
- Microwave devices
- Antenna devices
- Power devices
- Microelectronic devices and research
- Measurement equipment
- Circuital devices
- Radar systems
- Radiofrequency

Fig 3. - Old Style Naval Consolle © by Leonardo

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A. *Radiofrequency power devices*

Radar operations are based on the generation and modulation of energy at radiofrequency. In the thirties, the Magnetron, an oscillator that directly supplies a high-power signal, was introduce. Later, Klystron and TWT have been introduced and widely used as amplifiers. Since'70s, solid state components like transistors have replaced tube amplifiers. This implies that the required power level can be generated by arrays of elementary amplifiers, instead of single devices, increasing reliability and efficiency. This section contains exhibits like the H.V. reservoir voltage switch tube (1955), X-Band and ET828 KLYSTRON (1970), L-band solid state driver amplifier (1978), S-Band TWT (1980), L- band solid state power amplifier (1990) and S-Band solid state driver amplifier (1990). High Frequency devices for generation and processing of radar signals. In the beginning, radar systems could only detect the width of echo signals without separating real echoes from environmental clutter. Thanks to advance in signal processing, this problem is now completely solved. After thermo-ionic tubes followed by transistors working in high frequency, current technology uses integrated circuits and/or hybrids mounted on printed substrates with controlled impedance. Main exhibits are IF Amplifier (1960), 30 MHz Signal Generation RTN 10x (1967), COHO generator signal (1969), Signal generators 670 MHz and 700MHz (1970), 350 MHz (1972), Fast tuning assembly RTN 10x (1975), Pre-amplifier/mixer 700/30 MHz (1978), Quartz Oscillator RAT 31/Mr (1979), IF Amplifier with thermionic tubes (1980), Matched filter assembly (1983), 320 MHz amplifier and BITE R.F. (1990), IF double receiver with two matched channel (1999).

B. *Microwave devices*

To improve system accuracy and reduce antenna size at the same time, engineers used higher and higher transmission frequencies. Since 50's almost all radars operate in the microwaves. In this area, the development of semiconductors allowed manufacturing printed circuits with special substrate (e.g. alumina, RT-Duroid) able to elaborate HF signals. The relevant display case contains a Resonant cavity oscillator (1952), a Ferrite Modulator (1953), Cavity Frequency meter (1954), Waveguide double mixer and mixer orthomode (1960), Resonant cavity filter (1965), Waveguide pre-selector Filter (1970), X-band Stalo amplifier (1970), Sirio satellite Coaxial Diplexer (1971), Gunn Diode Amplifier (1976), C to X band frequency multiplier (1979), Waveguide diplexer (1980), Phase/ amplitude modulator (1980).

C. *Antenna Devices*

- The use of radio-waves to transmit and to receive information in range always needs transition structures to and from free space. One of these first structures is the "RUHMKORFF spark-coil". Antenna is generally both receiving and transmitting in the radar field (Figure 4). Engineers have always improved antennas with high gain, low sidelobes and with the possibility to electronically control the transmitting/receiving beam direction. The electromechanical radio-frequency components have been nowadays replaced by structures implemented through circuital techniques.

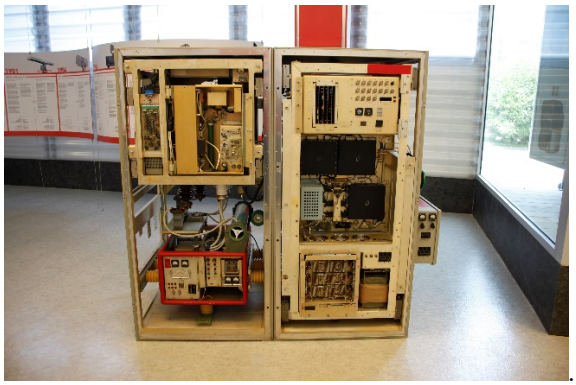

Figure 4 - Transmitter © by Leonardo

This section displays a Parabolic reflector (1953), Feed and Phase-Shifter for INTELSAT satellite antenna (1963), G33 Antenna Ceramic Feed (1963), Monopulse Feed (RTN 30X Antenna) (1968), Corrugated Feed for Satellite (1974), an ASPIDE missile Electroformed Monopulse Feed (1980) (Figure 5), Falco Radar Electroformed horn (1984), Double Dipole of the ALE 3X5 Antenna (1996) for air traffic control. At a side RAN 20S Antenna radiant column (1996), the RAT 31SL Antenna radiant row (1985) and the ALE 9 Antenna radiant column (1990) are also displayed.

Figure 5 -Missile Aspide Monopulse © by Leonardo

D. *Power Devices*

Power devices and systems are crucial components of any radar. They must satisfy very stringent requirements like accuracy of supplied voltages, residual noise and transient responses, in addition to their standard functions to supply the system with electrical power. For a radar system a wide range of voltage is necessary, from 5 Vdc to 40,000 Vdc. Currently, power systems consist in modular solutions based on COTS components, replacing power systems once designed for specific applications. Some examples are the HV Power Supply for A/R tube (1968), the 10 KV HV Power Supply for TWT (1976), the AC/DC converter module and a LV AD/DC Inverter (1970), the HV command for Tetrodo grid (1977) and a Grid voltage regulator circuit (-325 Vdc), (1980).

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E. Microelectronical Devices

The technological innovation in complex electronic products like radars has determined the maximum circuital integration. Currently research activities are focused on material (i.e. ceramics, ferromagnetic etc.) and on the manufacturing technologies for components made with these materials. Leonardo is strongly involved in this R&D activity, thanks to an-house modern microelectronic centre and a foundry for manufacturing e "GaAs" (Gallium Arsenide) components. Among the others, the related display case contains a Triple mixer on Alumina substrate – Working phase (1980), a Multilayer flexible printed circuit (1985), "SAW" Technology Delay Line (1986), Hybrid circuits – Product evolution (1975-1992), RT-DUROID "PIN" diode switch (1990), Diode phase-shifter (1992), CLAD R.F. distribution lines (1995) and some RTDUROID interdigital filters (1996).

F. Measurement equipment

Measurement equipment have always been fundamental to manufacture units and Radar systems, since they allow defining and measuring expected performances, guaranteeing stringent specifications. In order to achieve these goals, it is necessary to develop test stations with suitable software for managing several measurement tools.

G. Circuital Devices

The development of connections between parts and within them evolved side by side to other sectors of electronics. In the beginning, connections were many and point to point. Tubes in miniature, transistors and then integrated circuits called for the need to miniaturize connections between components mounted on planar structures, printed circuits, on which connections are carried out with a photoetching technique (Figure 6).

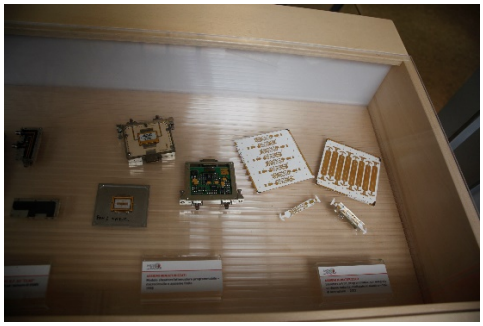

Figure 6 - Chips © by Leonardo

The complexity and the density of circuits are always increasing and consequently the density of connections arises. They need to use both faces of the layer and then to carry out connections on many layers which are glued to each other (multi-layer circuits, up to 20 layers). Radiofrequency devices represent a special case. The dimensions of connections and their tolerances are of fundamental importance together with the electrical and magnetic characteristics of the layers. The case contains some circuits of different periods, such as the 4 KB RAM Memory circuit (1970), a 36 KB RAM Memory circuit (1970) Single- layer circuit - "Sawtoothed" generator (1970-1975), the Single- layer circuit – ± 5/ +15 Vdc Voltage regulators (1976), Circuit of DDS 80 Display Unit (1980), Single-layer circuit – Timing generator (1981) and a Single layer circuit – HV Voltage regulators (1983).

H. Display devices

It is common opinion that the radar consists of two main parts, the antenna and the display to represent the received echo signals. From a design point of view, these systems didn't undergo significant changes, despite the technological evolution in the last years. Initially, the radar display had an oscilloscope shape, but soon it became round shaped with echoes represented in polar co-ordinates. This was the origin of the P.P.I (Plane Position Indicator), up to now basically unchanged even if using a P.C. monitor. This section presents a target detection panel (1965), the Vertical IPN 10 Single console (1978) and the Horizontal IPN 10 Tactical Console (1978) for naval systems.

I. The Radar Systems

Both inside the exhibition area and in the outside garden, the museum presents a suite of radar systems developed in the last fifty years. Some models are prototypes of radar systems that are now the best in class for technological development and leading products on the market. Like technology, also the radar systems on display are part of the history of radar in Italy, In particular Antenna AN/TPS 1: Search and Target Acquisition radar The AN/TPS-1 radar (Figure 7) is a medium power, L-band system designed for air target search and acquisition. Developed in 1954 to meet the severe requirements of the US marine Corps, for many years it has been the search and acquisition radar in NATO countries, representing their air defence backbone. AN/TPS is one of the first examples of transportable radar. Its most significant characteristic is the use of an MTI (Moving Target Indicator) with a range selectivity that enables the identification of air targets in case of intense fixed clutter.

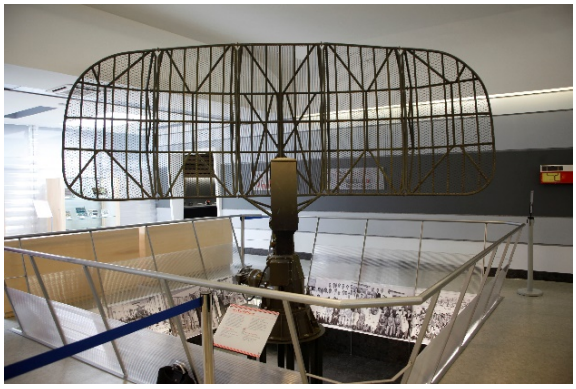

Figure 7 - The AN7TPSI Antenna © by Leonardo

1) EMPAR-Naval European Multifunctional Radar

The EMPAR (European Multifunction Phased Array Radar) has been the first multifunctional radar designed by the company in 2000 and one of the first worldwide. Installed onboard the Class Horizon new frigates and on Cavour aircraft-carrier, the EMPAR radar has continuously evolved to be integrated in any naval combat management system. EMPAR has also been adopted onboard the new FREMM (European Multi-Mission Frigate) frigates. The radar operates in C band, carrying out target detection, multiple tracking and missile guidance.

2) RAN 20S – Naval Search Radar

The RAN 20S radar is an S-band system with extended coverage in elevation, able to detect both air and surface targets. Developed at the end of the 90's, it is based on a

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solid-state transmitter. RAN 20S ensures the detection of air targets up to 65 Nautical Miles in range and up to 500 feet in altitude. It works both in long- and short-range configuration.

V. The Fusaro Plant hosting the museum

The choice of the Fusaro site stems from the history of the company and its activities in the development and provision of radar systems all over the world. The Fusaro site features about 70 years of experience within radar engineering. As a pioneer in this field, it has gone through the most important technological sector developments, actively contributing to the growth and innovative solutions development, as well as the international positioning of the brand. In Fusaro, over 1,500 radars have been manufactured for more than 150 countries worldwide: from the first systems based on an American-licensed analogical mechanic post-war technology, to the international success of the electronic phased-array antennas for the EMPAR shipboard radar, and the RAT 31 DL air defence radar provided to NATO member countries. Today, the Fusaro plant, part of the Leonardo Electronics Division, is dedicated to the manufacturing, production, integration and acceptance activities of the systems, including:

- Industrialization and manufacturing of ground, naval and ATC radar systems
- Software engineering
- Integration and testing
- Near Field area testing [anechoic room] and an external area to test radar.

VI. The company

Leonardo is a global industrial group, among the main players in Aerospace, Defence and Security that realises multi-domain technological capabilities in Helicopters, Aircraft, Aerostructures, Electronics, Cyber Security and Space. With over 60,468 employees worldwide, the company has a solid industrial presence in Italy, the UK, Poland and the US. It also operates in 150 countries through subsidiaries, joint ventures and investments. Innovation, continuous research, digital industry and sustainability are the pillars of its business worldwide. Leonardo has always fostered cultural initiatives of different kinds focused on the promotion, valorisation and protection of art, culture, history, and experience and traditions of its personnel both internally and towards its main stakeholders. Coordinated by the Fondazione Leonardo - ETS the new Integrated Museum and Archive System brings together the museums, historical archives and other exhibitions owned by Leonardo.

The “Museum System” comprises the Agusta Museum in Cascina Costa (VA), the Breda Meccanica Bresciana Museum in Brescia, the Museo delle Officine Galileo in Campi Bisenzio (FI), the Radar Museum in Bacoli (NA), the headquarters of the Museo della Melara Association in La Spezia and the new Aeronautical Industry Museum and the industrial archives of Agusta, Breda, OTO Melara, SIAI Marchetti, together with the Aermacchi Centre of Historical documents, Venegono (VA) and the Historical Documentation Centre in Turin and Caselle.

The aim of the 'Museum System' is to enhance and communicate the company's heritage in Italy, celebrating its historical, industrial and technological achievements. The wealth of material held in the archives is testament to the care and commitment shown by senior managers and volunteers who have worked tirelessly to preserve the company's collections over the years. The museums and archives recount the progress made over many decades which now underpins Leonardo’s standing as one of the world’s major players in aerospace, defence and security today.

The museums and archives, located adjacent to the production sites and housing a wide selection of technological collections, aim to be a cultural point of reference for the local communities in which they are located and for future generations. [10]

VII. Conclusions

Since its foundation, with the major aim to represent the company industrial heritage, the radar museum has now become a consolidated tool for disseminating technological culture among young students, as well as to strengthen the relationships between the company and its main stakeholders. In fact, every year the Leonardo Radar Museum is visited by several international and national Armed Forces delegations, Universities and schools, also hosting high level conferences.

Then the Radar Museum also contributes to the diffusion of scientific and technological matters among the young generations, attracting students towards scientific disciplines and then paving the way to the creation of new engineers, committed in new studies and developments. The Radar Museum while preserving the Italian radar history, also operates interacting actively with the surrounding communities, with cultural and scientific institutions as well with schools and universities.

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